Compact-Operator Algebras Determine the Underlying Banach Space

Candidate full solution to an explicit subquestion in arXiv:2505.02338
Model: GPT5.6

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Status: candidate full solution, likely valid. The compact-operator-algebra subquestion on page 40 of Guo–Wang is answered completely under the standard meaning of Banach-algebra isomorphism. The surrounding maximal Roe-algebra and K-theory questions are not claimed.

1 Source and exact target

Guo and Wang's paper [1] asks several questions about two maximal L^p-Roe completions. In the discussion following Question 8.1, on source PDF page 40, the authors write that it is unknown to them whether

$$\mathcal{K}(L^p(\mu)) \quad \text{is isomorphic to} \quad \mathcal{K}(\ell^p),$$

after noting the non-isomorphism of $L^p(0,1)$ and ℓ^p. We answer the concrete case $L^p(0,1)$, interpreting "isomorphic" in the standard Banach-algebra sense: a bounded linear bijective algebra homomorphism.

of the Rips complex to the Roe algebra of the Rips complex induces a canonical assembly map as the scale of the Rips complex tends to infinity. The maximal L^p-coarse Baum-Connes conjecture can be defined by claiming the assembly map is an isomorphism.

Notice that $L^p(0,1)$ is not isomorphic to ℓ^p and it is unknown to us whether the compact operator algebra $\mathcal{K}(L^p(\mu))$ is isomorphic to that of $\mathcal{K}(\ell^p)$. Thus, it is natural to ask whether the two completions above coincide, or have the same K-theory? And which one should be proper to define the maximal L^p-coarse Baum-Connes conjecture? By Yu's cutting and pasting argument [Yu97], can one prove that the K-theory of these two maximal L^p-versions of localization algebras is isomorphic to the K-homology group? As we proved in Section 4, when $p \in (1,\infty)$, one can find Kazhdan projections in both $C^p_{\max,\mathscr{L}^p}(X)$ and $C^p_{\max,\ell^p}(X)$ for spaces with geometric property $(T_{\mathscr{L}^p})$. It is proved in [CN23] that expanders are counterexamples to the L^p-coarse Baum-Connes conjecture. Then, can the existence of Kazhdan projections provide counterexamples to the maximal version of the L^p-coarse Baum-Connes conjecture, as in [WY12b]?

Figure 1: The source paragraph on PDF page 40. The relevant compact-operator algebra sentence is the first complete sentence in the second paragraph.

2 Proof intuition

The algebra $\mathcal{K}(X)$ already contains a canonical copy of X, up to isomorphism. Choose a rank-one idempotent $P = x_0 \otimes x_0^*$. The left ideal $\mathcal{K}(X)P$ consists exactly of the rank-one operators $x \otimes x_0^*$ and is therefore isomorphic to X. It is algebraically a minimal nonzero left ideal. An algebra isomorphism sends it to a minimal left ideal $\mathcal{K}(Y)Q$. Since Q is a compact idempotent, it has finite rank; minimality forces that rank to be one. Thus the image ideal is isomorphic to Y, recovering $X \cong Y$.

For the L^p application, $L^p(0,1) \not\cong \ell^p$ when $p \neq 2$. At $p = 1$ the Schur property separates them. For $1 < p < \infty$, the Rademacher functions span a complemented copy of ℓ^2 in $L^p(0,1)$, while Pitt's compactness theorem prevents the corresponding embedding or complementation inside ℓ^p unless $p = 2$.

3 The algebraic rigidity theorem

For Banach spaces X, Y, write $\mathcal{K}(X)$ for the Banach algebra of compact operators on X with the operator norm.

Theorem 1 (compact-operator algebra rigidity)**.** *Let X and Y be nonzero Banach spaces over the same scalar field. Then*

$$\mathcal{K}(X) \cong \mathcal{K}(Y) \quad \textit{as Banach algebras} \qquad \Longleftrightarrow \qquad X \cong Y \quad \textit{as Banach spaces.}$$

No approximation-property assumption is required.

Proof. Suppose first that $S : X \to Y$ is a Banach-space isomorphism. Conjugation,

$$T \longmapsto STS^{-1},$$

is a bounded Banach-algebra isomorphism from $\mathcal{K}(X)$ onto $\mathcal{K}(Y)$.

Conversely, let $\Phi : \mathcal{K}(X) \to \mathcal{K}(Y)$ be a bounded Banach-algebra isomorphism. Choose $x_0 \in X$ and $x_0^* \in X^*$ with $x_0^*(x_0) = 1$, and put

$$P = x_0 \otimes x_0^*, \qquad P(x) = x_0^*(x)x_0.$$

Then P is a rank-one idempotent. Its generated left ideal is

$$\mathcal{K}(X)P = \{x \otimes x_0^* : x \in X\}.$$

Indeed, $TP = (Tx_0) \otimes x_0^*$; conversely, every x occurs as Tx_0 for a rank-one operator T. The map

$$X \longrightarrow \mathcal{K}(X)P, \qquad x \longmapsto x \otimes x_0^*,$$

is a Banach-space isomorphism, since $\|x \otimes x_0^*\| = \|x\| \, \|x_0^*\|$.

The left ideal $\mathcal{K}(X)P$ is algebraically minimal. To see this, take a nonzero $x \otimes x_0^*$ in a left subideal. Given any $z \in X$, a rank-one operator A can be chosen with $Ax = z$, so $A(x \otimes x_0^*) = z \otimes x_0^*$ also lies in that subideal.

Set $Q = \Phi(P)$. Then Q is a nonzero compact idempotent and

$$\Phi(\mathcal{K}(X)P) = \mathcal{K}(Y)Q.$$

Thus $\mathcal{K}(Y)Q$ is a minimal nonzero left ideal. The range of a compact idempotent is finite-dimensional: the idempotent restricts to the identity on its range, and a compact identity can occur only in finite dimension. If $\operatorname{rank} Q > 1$, choose a rank-one idempotent R whose range lies in $\operatorname{ran} Q$ and such that $RQ = R$. Then

$$0 \neq \mathcal{K}(Y)R \subsetneq \mathcal{K}(Y)Q,$$

where properness follows because equality would express $Q = AR$ for some A, forcing $\operatorname{rank} Q \leq 1$. This contradicts minimality. Therefore Q has rank one.

Writing $Q = y_0 \otimes y_0^*$ with $y_0^*(y_0) = 1$, the same argument as above identifies $\mathcal{K}(Y)Q$ with Y. The bounded restriction of Φ, with its bounded inverse, now gives

$$X \cong \mathcal{K}(X)P \cong \mathcal{K}(Y)Q \cong Y.$$

$\square$

4 Application to $L^p(0,1)$ and ℓ^p

Lemma 2. *For $1 \le p < \infty$,*

$$L^p(0,1) \cong \ell^p \qquad \iff \qquad p = 2.$$

Proof. For $p = 2$, both spaces are separable infinite-dimensional Hilbert spaces and are isometrically isomorphic.

For $p = 1$, ℓ^1 has the Schur property, whereas $L^1(0,1)$ does not. In fact, the Rademacher functions r_n satisfy $\|r_n\|_1 = 1$ and converge weakly to zero: for $g \in L^\infty(0,1) \subset L^2(0,1)$, the coefficients $\int g r_n$ tend to zero by Bessel's inequality.

Let $1 < p < \infty$. Khintchine's inequality [2] shows that the closed Rademacher span

$$R_p = \overline{\operatorname{span}}\{r_n : n \ge 1\} \subset L^p(0,1)$$

is isomorphic to ℓ^2. It is complemented. Indeed, for finite scalar sequences (b_n), Hölder's and Khintchine's inequalities give

$$\left| \sum_n b_n \int f r_n \right| \le \|f\|_p \left\| \sum_n b_n r_n \right\|_{p'} \le C_{p'} \|f\|_p \|(b_n)\|_2.$$

Thus $(\int f r_n)_n \in \ell^2$ with norm controlled by $\|f\|_p$, and another application of Khintchine's inequality makes

$$f \longmapsto \sum_{n=1}^{\infty} \left(\int_0^1 f r_n \right) r_n$$

a bounded projection onto R_p.

Pitt's theorem [3] says that every bounded operator $\ell^s \to \ell^r$ is compact when $1 \le r < s < \infty$. If $1 < p < 2$, an embedding $\ell^2 \hookrightarrow \ell^p$ would itself be compact by Pitt, which is impossible for an isomorphic embedding of an infinite-dimensional Banach space. If $2 < p < \infty$ and ℓ^p contained a complemented copy of ℓ^2, an embedding $J : \ell^2 \to \ell^p$ and projection onto $J\ell^2$ would yield a bounded operator $S : \ell^p \to \ell^2$ with $SJ = I_{\ell^2}$. Pitt makes S compact, hence makes the identity compact, again impossible. Therefore ℓ^p cannot be isomorphic to $L^p(0,1)$ for $p \ne 2$. □

Corollary 3 (answer to the source subquestion)**.** *For $1 \le p < \infty$,*

$$\boxed{\mathcal{K}(L^p(0,1)) \cong \mathcal{K}(\ell^p) \text{ as Banach algebras} \quad \iff \quad p = 2.}$$

In particular, the answer is negative for every $p \ne 2$ and affirmative for $p = 2$.

Proof. Combine Theorem 1 with Lemma 2. □

5 Verification, limitations, and novelty check

Verification. The proof was checked directly at each algebraic step. The only external inputs in Lemma 2 are Khintchine's inequality and Pitt's theorem; the complemented Rademacher projection is derived explicitly. The theorem uses neither a basis nor an approximation property. The rank argument also checks that a minimal left ideal generated by a compact idempotent must come from a rank-one idempotent.

Scope. The conclusion concerns bounded Banach-algebra isomorphisms. It does not decide whether the two compact-operator spaces are isomorphic merely as Banach spaces. It also does not decide whether the two maximal L^p-Roe completions in the source coincide or have the same K-theory. Finally, for a general measure μ, the answer depends on the Banach-space isomorphism type of $L^p(\mu)$; the displayed corollary is for the nonatomic model $L^p(0,1)$ explicitly named by the source.

Novelty search. On 19 July 2026, bounded searches covered the run's registry, solution, attempt, and proof-gap indexes; exact arXiv-id/title searches; and web/arXiv queries for variants of "$\mathcal{K}(X)$ isomorphic to $\mathcal{K}(Y)$", compact-operator algebras and minimal left ideals, and "$\mathcal{K}(L^p)$ isomorphic to $\mathcal{K}(\ell^p)$". No source stating this exact answer was located. The minimal-left-ideal mechanism is elementary and may well be standard ring-theoretic folklore, so originality confidence is conservative. The mathematical claim does not depend on novelty.

Human review recommendation. Check that "isomorphic" in the source paragraph was intended as Banach-algebra isomorphism rather than Banach-space isomorphism. Under that standard algebraic interpretation, the proof is short and the result should be promoted after review.

References

[1] L. Guo and Q. Wang, *Geometric Banach property (T) for metric spaces via Banach representations of Roe algebras*, arXiv:2505.02338v2 (2025).

[2] A. Khintchine, *Über dyadische Brüche*, Math. Z. **18** (1923), 109–116.

[3] H. R. Pitt, *A note on bilinear forms*, J. London Math. Soc. **s1-11** (1936), 174–180.